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Maintenance and management of civil infrastructure based on condition, safety, optimization, and life-cycle cost*

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Cost-competent maintenance and management of civil infrastructure requires balanced consideration of both the structure performance and the total cost accrued over the entire life-cycle. Most existing maintenance and management systems are developed on the basis of life-cycle cost minimization only. The single maintenance and management solution thus obtained, however, does not necessarily result in satisfactory long-term structure performance. Another concern is that the structure performance is usually described by the visual inspection-based structure condition states. The actual structure safety level, however, has not been explicitly or adequately considered in determining maintenance management decisions. This paper reviews the recent development of life-cycle maintenance and management planning for deteriorating civil infrastructure with emphasis on bridges using optimization techniques and considering simultaneously multiple and often competing criteria in terms of condition, safety and life-cycle cost. This multiple-objective approach leads to a large pool of alternative maintenance and management solutions that helps active decision-making by choosing a compromise solution of preferably balancing structure performance and life-cycle cost.

Keywords: Civil Infrastructure; Maintenance; Life-cycle cost; Performance; Safety; Multiobjective optimization; Uncertainty

1. Introduction

Satisfactory lifetime performance of civil infrastructure is of critical importance to sustained economic growth and social development of a modern society. In particular, the highway transportation system is considered to be one of the society's critical foundations. Among the many elements of this complex system, bridges are especially important and the most vulnerable because of their distinct function of joining highways as the crucial nodes. The bridge infrastructure throughout the United States, especially in the dense urban regions of the country, has been

constantly exposed to aggressive environments. It faces continuously increasing traffic volumes and heavier truck-loads, which degrade the long-term performance of highway structures at an alarming rate. In the United States, 27.1% of the 590,750 existing bridges nationwide are labeled structurally deficient or functionally obsolete; these bridges must be replaced or upgraded immediately with an estimated annual investment of US\$ 9.4 billion required for the next 20 years in order to eliminate all bridge deficiencies (ASCE 2005).

A deteriorating public works infrastructure leads to increased costs for business and ultimately for consumers.

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Catastrophic failure of civil infrastructure can cause widespread social and economic consequences. Nowadays, the homeland security has become a heightened concern (FHWA 2003). Well-maintained civil infrastructure can substantially increase a country's competitiveness in a global economy and enhance resilience to adverse circumstances such as natural hazards (e.g. earthquakes, hurricanes and floods) and manmade disasters (e.g. vehicular collision and explosive blasts due to terrorists' attacks). In addition to developing more advanced maintenance technologies, structure managers urgently need methodologies to cost-effectively allocate limited budgets for maintaining and managing aging and deteriorating civil infrastructure over the specified time period, with the goal of optimally balancing lifetime structure performance and whole-life maintenance cost.

Researchers and engineers around the world have been developing and implementing different maintenance management programs to achieve desirable solutions to maintain satisfactory infrastructure performance from the long-term economical point of view. In particular, most existing methodologies for the maintenance and management of the highway transportation infrastructure are based on least life-cycle cost while enforcing relevant performance thresholds. There exist practical difficulties with these treatments. For example, if the available budgets are more than the computed minimum life-cycle cost, the infrastructure performance can be maintained at a higher level than what was previously prescribed for deriving the minimum life-cycle cost solution. On the other hand, if the available financial resources are not enough to meet the computed minimum life-cycle cost, bridge managers have to accept a solution that can improve the structure performance to the highest possible level under the budget constraints. Indeed an overall satisfactory maintenance planning solution should therefore be determined by optimally balancing competing objectives of, for example, improving structure performance and reducing the long-term cumulative cost.

Another problem regarding the existing maintenance management systems is that visual inspection-based infrastructure condition states are mostly used to describe the infrastructure performance. The actual infrastructure safety level, however, has not been explicitly or adequately accounted for. Maintenance management decisions made on the basis of infrastructure condition states alone are not necessarily cost-effective and, in some cases, may even cause serious safety consequences. For example, it is possible that a reinforced concrete structure with satisfactory visual condition states could actually be structurally unsafe when invisible flaws such as corrosion of the embedded reinforcement exist. Failure to identify these facts definitely raises severe safety concerns. It is also possible that a structure may be structurally sound despite the poor visual condition status. Under this situation, only minor or moderate repair

of these visual defects may be necessary and thus costly upgrading or total replacement can be avoided. Savings of maintenance funds can then be enormous.

In response to the above concerns, all merit metrics relevant to civil infrastructure performance need to be considered *simultaneously* to determine the maintenance management solutions. The life-cycle cost is one such merit measure. Structure performance in terms of condition and safety states should also be treated as merit measures. Traditional methodologies treat the prescribed levels of these performance measures as constraints that valid maintenance management solutions should comply with. However, the advanced methodologies based on a multiobjective optimization formulation treat all these performance measures as additional merit objective functions that are not restrictive in nature and the actual performance levels are at the discretion of civil infrastructure managers. As a result, the present performance-based maintenance management methodologies lead to a group of non-dominated solutions, each of which represents a unique optimized tradeoff between life-cycle cost minimization and infrastructure performance maximization. That is, there does not exist another solution that can improve one objective of the optimized solution being considered without sacrificing at least one of the other objectives. This large set of alternative solutions helps civil infrastructure managers' active decision-making by selecting a maintenance solution with the most desirable balance between the conflicting life-cycle cost and infrastructure performance objectives. It is clear that this will have significant impacts on developing the next-generation of civil infrastructure maintenance management systems, which should integrate the concept of structure safety and reliability as well as consider other relevant conflicting objectives in order to meet maintenance managers' specified requirements and budget constraints.

Research on the use of multiobjective optimization techniques in maintenance management of civil infrastructure has appeared in the recent literature. Multiple and conflicting performance indicators such as condition, safety and durability along with life-cycle cost are simultaneously considered as separate criteria (Liu *et al.* 1997, Miyamoto *et al.* 2000, Furuta *et al.* 2004). Interestingly all these research efforts use genetic algorithms (GAs) as numerical optimization tools. This is because the practical maintenance management problems can be best posed as combinatorial optimization problems. For example, maintenance managers often require a list of prioritized maintenance interventions for civil infrastructure on an annual basis (Das 1999). Due to their inherent features (as will be discussed in a later section), GAs are very effective for solving these kinds of problems.

In order to make rational decisions on preserving deteriorating civil infrastructure, maintenance engineers need to appropriately address sources of uncertainty

associated with the deterioration processes of civil infrastructure under no maintenance and under different maintenance interventions. Sources of uncertainty include imperfect description of the mechanical loadings and environmental stressors as well as the inexact prediction of the deteriorating structure performance. There are two general types of uncertainty (Wen *et al.* 2003): aleatory and epistemic. The aleatory uncertainty is caused by the inherent variation of the deterioration due to the combined effects of complex traffic loadings and environmental stresses as well as structural aging. The epistemic uncertainty stems from the randomness caused by subjective assumption in evaluating demand and load-carrying capacity of structures or insufficient knowledge in understanding, for example, the deterioration mechanisms. This type of uncertainty can be lessened through, for example, Bayesian updating, when additional information is gathered (Enright and Frangopol 1999, Estes *et al.* 2004, Rafiq *et al.* 2004). The ensuing maintenance interventions over the specified time period of deteriorating structures add additional uncertainty in the prediction of time-dependent structure performance.

In this paper, the significance of using multiobjective optimization techniques in civil infrastructure maintenance management is emphasized. In particular, the recent development of life-cycle maintenance and management for deteriorating highway bridge infrastructure is discussed, in which the combinatorial optimization is used and multiple and competing criteria in terms of condition, safety and life-cycle cost are considered simultaneously. Uncertainties associated with the deterioration process under no maintenance and under maintenance interventions are treated by Monte Carlo simulation, which produces sample mean predictions of structure performances as well as of life-cycle maintenance cost. As numerical examples, GA-based automated maintenance management procedures are applied to prioritize maintenance resources for deteriorating structures over designated time periods. The first example deals with project-level maintenance management of preserving a population of similar highway reinforced concrete crossheads subject to various degrees of deterioration. The second example discusses the network-level bridge maintenance management, which considers individual bridges acting as the nodes of a highway transportation network based on network analysis under uncertainty. Finally, the research needs of integrating structural health monitoring with maintenance management systems are briefly discussed.

2. Performance and deterioration of civil infrastructure

2.1 Performance indicators

Appropriate performance indicators are needed to describe the time-dependent civil infrastructure performance, based

on which maintenance management decisions can be made. Some of the commonly used performance indicators are discussed as follows.

2.1.1 Condition index. Visual inspection-based condition rating index is traditionally used to measure the bridge's remaining load-carrying capacity (Pontis 2001). The current bridge management systems (BMSs) characterize structural elements by discrete condition states noting deterioration (Thompson *et al.* 1998). Typically maintenance interventions should be prioritized to civil infrastructure with unacceptable and poor condition rating levels. For reinforced concrete elements under corrosion attack in the United Kingdom, Denton (2002) classifies the visual inspection-based condition states into four discrete levels, denoted as 0, 1, 2 and 3, that represent no chloride contamination, onset of corrosion, onset of cracking and loose concrete/significant delamination, respectively. A value larger than 3 indicates an unacceptable condition state. As a subjective measure, however, the condition index may not faithfully reflect the true load-carrying capacity of structural members.

2.1.2 Safety index. According to bridge specifications in the United Kingdom, the safety index is defined as the ratio of available to required live load capacity (DB12/01 2001). The structure performance is considered unacceptable if the value of safety index drops below 0.91.

2.1.3 Reliability index. The reliability index is used to quantify the structure safety level and can thus be used as a performance indicator (Melchers 1999, Nowak and Collins 2000). This is a more objective means to assess the performance of an existing structure, provided the probabilistic nature of structural load and capability can be properly modeled. For example, Estes and Frangopol (1999) evaluated the system reliability of a bridge structure that comprises both super- and sub-structure and is modeled as a series-parallel system. Using the reliability concept, Ghosn and Johnson (2000) assess the safety level of bridges subject to flood scour and seismic loads. Akgül and Frangopol (2003) conducted reliability analysis of bridge components using performance limit state functions defined in terms of standard code formulations in AASHTO (1996) specifications.

2.2 Deterioration mechanism and modeling

Accurate modeling of the structure deterioration process is the most critical component in maintenance management of civil infrastructure, because allocation of maintenance needs must be determined by predicting performance deterioration of civil infrastructure under no maintenance and under different maintenance scenarios. In doing so, it is

essential to understand the mechanism of structure deterioration.

2.2.1 Deterioration mechanism. The civil infrastructure deterioration is a very complex process and is affected by many factors, which concern physical, chemical and electrical phenomena, among others. For highway infrastructure such as bridges and roads, aggressive environmental conditions, including alkali-silica reaction, chloride contamination, and sulfate attack contribute to progressive performance deterioration over time. The ever-increasing traffic load effects further aggravate the structure performance. It should be noted that the identified mechanisms cannot explain the entire performance deterioration. Uncertainties are unavoidable in the understanding and proper mathematical treatment of the deterioration process.

2.2.2 Deterioration modeling. The deterioration process of civil infrastructure has been widely depicted using Markov chain models. This mathematical model is built under the premise that current states are only dependent upon a finite number of previous states. If the current state is only a function of its immediately past state, it is called a one-step Markov chain model. The core element of a Markov chain model is a transition matrix that dictates the probability that one state component changes to another state. In most practical cases for maintenance management of civil infrastructure, a stationary Markov chain model is used in which the transition matrix remains the same throughout the specified time period.

The time-dependent performance deterioration of civil infrastructure under no maintenance and under maintenance interventions can also be predicted by continuous computational models. For example, Frangopol *et al.* (2001) proposed a model to describe the deterioration process under no maintenance by a function characterized by an initial performance level, time to damage initiation, and a deterioration curve governed by appropriate functions and in the simplest form, a linear function with a constant deterioration rate. Effects of a generic maintenance action include immediate performance enhancement, suppression of the deterioration process for a specified time interval, reduction in the deterioration rate, and another specified duration of maintenance effect beyond which the deterioration rate resumes to the original one. Uncertainties associated with the deterioration process are considered in terms of probabilistic distribution of the controlling parameters of this computational model. Monte Carlo simulation is then used to account for these uncertainties by obtaining the statistical time-varying performance profiles of deteriorating structures under various single-cyclic and combined maintenance actions.

The performance deterioration may also be predicted if reasonable mechanistic models are available. For example,

the deterioration of the reinforcement in reinforced concrete structures is usually caused by de-icing chemicals related corrosion, whose initiation in concrete can be reasonably predicted by Fick's law of diffusion (Thoft-Christensen 1998). Effects of corrosion are considered as a gradual reduction of the reinforcement area. The time-dependent structural capacity can then be predicted. If the information on time-variant loads is available, the structural reliability profiles can be obtained as a result. For example, Enright and Frangopol (1999), Val *et al.* (2000) and Akgül and Frangopol (2003) predict the time-dependent reliability deterioration of highway bridges. Due to the existence of uncertainty in initiation time and rate of corrosion as well as other relevant parameters, Monte Carlo simulations are needed to account for such uncertainty effects on the prediction of deteriorating performance.

3. Numerical optimization for maintenance management

Maintenance management can be readily formulated as a combinatorial optimization problem because there always exist objectives to be optimized. The objectives can be life-cycle cost minimization under structure performance constraints, structure performance maximization under budget constraints, and simultaneous consideration of reducing life-cycle cost and increasing structure performance. Clearly the last criterion is the generalization of the first two objectives. As discussed previously, this paper is focused on the multiobjective optimization formulation. Therefore, effective numerical optimization tools for this formulation are needed.

3.1 Traditional algorithms

Most traditional optimization algorithms including mathematical programming are problem-dependent. They usually make use of gradient information to guide the search process and often assume continuous-valued design variables. This may lead to significant difficulties when design variables can only take discrete values such as in the present maintenance management problems. More importantly, most traditional methods by themselves can only handle single-objective based optimization problems, for which a single final optimal solution is sought. In order to solve optimization problems with multiple and usually conflicting objective functions, one has to convert, with non-trivial inconvenience, the original multiobjective problem into a series of equivalent single-objective optimization problems, which in total produce a set of solutions that displays optimal tradeoff among all competing objectives. There are two basic approaches commonly used for this treatment. One approach is the ϵ -constraint method, which keeps only one objective at a time and

converts the other objectives into constraints. The difficulty associated with this method lies in the determination of constraint values, especially the upper and lower limits. The other approach is the weighted sum method, which forms a single composite objective function as a weighted sum of the original multiple objectives, using a specified set of weight coefficients. Because traditional optimization methods usually search for the optimized solution on a point-by-point basis, multiple algorithm runs are needed, each at best generating one of those optimized tradeoff solutions.

3.2 Heuristic algorithms

Genetic algorithms (GAs) as one type of heuristic algorithms are stochastic search and optimization tools that follow the survival-of-the-fittest principle from the biological sciences (Goldberg 1989). Since they formally appeared in the 1960s, GAs have been successfully applied to a wide range of problems because of their ease of implementation and robust performance. The population-based GAs are general-purpose numerical tools. Gradients are no longer needed and discrete-valued design variables can be handled without any difficulty. This is particularly useful for the present maintenance planning problems where different maintenance actions have to be scheduled at discrete years. More importantly, GAs can handle multiple conflicting objectives directly and simultaneously. Because GAs maintain many solutions at the same time, a distribution of optimized tradeoff solutions can be obtained with a single algorithm run.

A successful multiobjective optimization algorithm should obtain a non-dominated set of solutions close to the global Pareto optimal front and have this solution set as diverse as possible in order to best possibly prevent solution clustering. Unlike single-objective problems where the objective function itself may be used as the fitness measure to rank the merit of individual solutions, multiobjective GAs need a single fitness measure that reflects the overall merit of multiple competing objectives (Deb 2001). Many of the existing multiobjective GAs adopt the non-dominated sorting technique (Goldberg 1989) to rank all solutions in a population based on the concept of domination.

4. Life-cycle cost oriented design and maintenance management of civil infrastructure

4.1 Overview

Consideration of lifetime expenses during the initial design stage of civil infrastructure has received fruitful research attention in recent years. A life-cycle cost analysis consists of calculation of not only the initial construction cost, but also costs due to operation, inspection, maintenance,

repair, and damage/failure consequences during a specified lifetime. Future costs are usually discounted to the present value for meaningful comparison. The goal is to produce a cost-effective design solution that balances the initial cost and lifetime cost in the preferred manner. Chang and Shinozuka (1996) reviewed the life-cycle cost analysis for civil infrastructure with emphasis on natural hazard risk mitigation. Hassanain and Loov (2003) discussed cost optimization of concrete bridge components and systems and review developments in life-cycle cost analysis and design of concrete bridges.

The minimum expected life-cycle cost has been the most widely used criterion in design optimization of new structural systems considering lifetime performance. The general form of the expected life-cycle cost can be calculated as (Frangopol *et al.* 1997):

$$C_{ET} = C_T + C_{PM} + C_{INS} + C_{REP} + C_F \quad (1)$$

where C_{ET} = expected total cost, C_T = initial design/construction cost, C_{PM} = expected cost of routine maintenance, C_{INS} = expected cost of performing inspections, C_{REP} = expected cost of repairs and C_F = expected cost of failure. Application of expected life-cycle cost minimization has been made for the design of both bridges and buildings. Note that for maintenance planning of existing structures over a prescribed time period, the expected total intervention cost, which comprises all but the initial cost, needs to be used instead.

4.2 Application to design of bridges and buildings

Frangopol *et al.* (1997) proposed a conceptual framework for reliability-based life-cycle cost optimal design of deteriorating reinforced concrete bridges considering inspection and repair planning, based on the following formulation:

$$\text{Minimize } C_{ET} \text{ subject to } P_{f,life} \leq P_{f,life}^* \quad (2)$$

where $P_{f,life}$ and $P_{f,life}^*$ are the computed and acceptable lifetime failure probabilities, respectively. The reliability-based lifetime approach considers the quality of various inspection techniques, all repair possibilities based on an event tree, effects of aging and corrosion deterioration, damage intensity, effects of repair on structural reliability, and the time value of money. It is pointed out that the quantification of uncertainties in the input data, including the rate of corrosion, quality of inspection methods and failure cost, is necessary for obtaining a reliable structural design integrated with economic and safety considerations. Bayesian updating of damage detection probabilities, use of improved time-varying bridge reliability models, and selection of realistic target reliability levels are all needed.

The studies by Frangopol *et al.* (1997, 2000, 2001) are concerned with designing or maintaining individual bridges. It is known that bridges represent the most critical components of highway networks. Not only under disastrous natural hazards such as flood and earthquakes but also under daily traffic conditions, malfunctioning of bridges could likely disrupt the normal transportation. This causes significant problems to the welfare of the entire society. From this point of view, it is more important to maintain the normal functionality as well as extend life expectancy of the bridge network as a whole instead of dealing with each bridge separately. Therefore, life-cycle cost oriented design and maintenance optimization of all bridges within a network poses a natural extension of the previous studies and deserves future research (Augusti *et al.* 1994, 1998, Akgül and Frangopol 2003). When analyzing the bridge network, user costs in terms of, for example, travel delay and additional vehicle fuel consumption as well as accidents that are caused by reduced serviceability of a bridge network due to maintenance and/or inadequate capacity, need also to be addressed (Chang and Shinozuka 1996).

Recent earthquakes reveal that economic losses caused by less drastic structural damages of building structures as well as ensuing functional disruptions could be comparable to the structure's initial cost. A life-cycle cost analysis provides a convenient means for decision makers to compare seismic design alternatives using economic terms. Both initial construction expenses and lifetime cost due to seismic damage/failure consequences as well as associated repair and retrofitting efforts need to be properly addressed in life-cycle cost analysis. A cost-effective building design should appropriately balance initial and future costs. The minimum expected life-cycle cost has been the most widely used criterion for the seismic design of steel (Kang and Wen 2000) and reinforced concrete buildings (Ang and Lee 2001). A group of conventional trial structural designs with different seismic capacities are investigated and no automated numerical optimization is performed. Cheng and Ang (1999) suggested a reliability-based optimization procedure using GA for cost-effective design and upgrading of seismic-resistant reinforced concrete building structures; the target reliability and the minimum expected life-cycle cost are selected as the objective functions.

Liu *et al.* (2004) proposed a multiobjective optimization approach for solving life-cycle cost oriented seismic design of steel moment frame structures. Initial cost and lifetime seismic damage cost are considered as separate objective functions subject to simultaneous minimization. The lifetime seismic damage cost is computed as (Kang and Wen 2000):

$$C_{seismic} = (C_1P_1 + C_2P_2 + \cdots + C_jP_j + \cdots + C_NP_N) \frac{\nu}{\lambda} (1 - e^{-\lambda T}) \quad (3)$$

where N = number of seismic damage states considered; C_j = cost function of the j -th seismic damage state, which consists of direct structural/nonstructural damage and repair cost, cost due to loss of contents, relocation cost, direct/indirect economic loss, and human injury and fatality cost; P_j = j -th damage state probability given seismic occurrence; ν = annual occurrence rate of major seismic events modeled by a Poisson process; λ = annual monetary discount rate; T = service life. This formula implicitly assumes that a damaged structure is immediately retrofitted to its original intact condition after each damage-inducing seismic event. Seven damage states are defined in terms of maximum interstory drift ratio as the seismic performance index. Effects of randomness and uncertainty in estimating seismic demand and capacity as well as in describing seismic hazards are considered in accordance with SAC/FEMA guidelines (BSSC 2000).

4.3 Application to bridge maintenance management

The life-cycle cost analysis provides a rational approach to cost-effective prioritization under budget constraints for maintaining satisfactory lifetime performance of deteriorating highway bridges. In the United States, bridge management systems (BMSs), such as Pontis (Thompson *et al.* 1998) and BRIDGIT (Hawk and Small 1998), are used by state departments of transportation. As stated previously, these BMSs are built upon visual inspection based discrete condition states and Markovian deterioration modeling using stationary transition probabilities; lifetime reliability concepts are usually not directly incorporated. These inherent deficiencies cause limitations of these existing BMSs (Frangopol *et al.* 2001).

As an advance over these BMSs, some researchers proposed reliability-based life-cycle maintenance management using optimization (e.g. Frangopol and Das 1999, Thoft-Christensen 1999). Frangopol (1999) presented a systematic framework for cost-effective life-cycle bridge management that integrates both bridge lifetime reliability and life-cycle cost analysis. A review on these subjects can be found in the literature (Frangopol *et al.* 2001). The recently published NCHRP Report 483 'Bridge life-cycle cost analysis' of NCHRP Project 12-43 (Hawk 2003) provides guidelines for transportation agencies to conduct life-cycle cost analysis of major bridges. Representative studies done by the first author and his former graduate students include: (a) Frangopol *et al.* (2000) present realistic examples of optimal bridge maintenance planning based on minimum expected life-cycle cost criterion; and (b) Kong and Frangopol (2003) use a modified event tree analysis to compute probabilities of maintenance actions and the expected life-cycle cost of deteriorating highway bridges.

5. Maintenance management at the project level

5.1 Overview

Tremendous efforts have been devoted by civil engineering researchers and practitioners to develop methodologies and techniques for long-term maintenance management of deteriorating civil infrastructures, in particular bridges. Most of the previous research on bridge maintenance management can be categorized as project-level based because only individual deteriorating bridges or a group of similar bridges are dealt with. Many of the existing maintenance planning optimization procedures rely on deterministic evaluation of bridge performance only. Uncertainties, however, inevitably exist in describing the time-dependent deterioration process under no maintenance as well as under different maintenance interventions. A life-cycle cost analysis would be questionable if constraints are imposed on relevant bridge performance for which the associated sources of uncertainty are not taken into account. In order to carry out a realistic life-cycle cost analysis, uncertainties have to be considered in predicting lifetime bridge performance. As previously indicated, both aleatory and epistemic uncertainties have to be considered in predicting lifetime bridge performance.

5.2 Numerical example

5.2.1 Problem statement. In the first numerical example, a GA-based procedure is used to prioritize maintenance needs for deteriorating reinforced concrete highway crossheads through simultaneous optimization of both structure performance (in terms of condition and safety states as defined in sections 2.1.1 and 2.1.2) and life-cycle maintenance cost with a prescribed discount rate. The maintenance management problem is thus posed as a combinatorial multiobjective optimization problem in that, for any year over the specified time period, at most one of the five different maintenance strategies (as defined in the next subsection) may be carried out. The optimization problem is conceptually stated as:

Goal Obtain a set of tradeoff lists of maintenance actions applied over the specified time period that:

Objectives

- minimize the largest (i.e. worst) lifetime condition index value;
- maximize the smallest (i.e. worst) lifetime safety index value; and
- minimize the present value of life-cycle maintenance cost.

Subject to

- condition index value ≤ 3.0 ; and
- safety index value ≥ 0.91 .

Design variables of the present optimization problem are discrete years and types of maintenance applications. A generic maintenance solution is then a list of maintenance interventions prioritized on an annual basis. Modeling of the structure deterioration processes under no maintenance and under different maintenances follows Frangopol *et al.* (2001); uncertainties associated with this modeling are accounted for by means of Monte Carlo simulation.

5.2.2 Maintenance strategies. The condition and safety deterioration processes of a large population of reinforced concrete crossheads under no maintenance are characterized with the data provided elsewhere (Denton 2002, Liu and Frangopol 2005b); information of the five maintenance strategies used in this numerical example, i.e. replacement of expansion joints, silane, cathodic protection, minor concrete repair and 'do nothing but rebuild' is also given. All random variables are assumed to be triangularly distributed and, consequently, defined in terms of minimum, mode and maximum values, respectively. Replacement of expansion joints is statistically the least costly maintenance strategy. It does not improve performance or delay deterioration but alleviates the severity of deterioration of both condition and safety performance. The silane treatment reduces chloride penetration but does not correct existing defects or replace deteriorated structural components. Silane does not improve performance or delay deterioration either. Statistically, silane reduces the deterioration of the condition more efficiently than replacing the expansion joints while having the same effects on safety deterioration. However, silane usually costs more than replacing the expansion joints. Cathodic protection replaces anodes and thus suppresses the corrosion of the reinforcing bars almost completely. It postpones the deterioration of both condition and safety for a relatively long time period (12.5 years) upon application. The minor concrete repair strategy is applied to replace all cover concrete with visual defects but not corroded reinforcing bars. It significantly improves the condition performance while the deterioration in safety is delayed until the condition index exceeds 1.0. Finally, the 'do nothing but rebuild' strategy increases both condition and safety levels to those values typical of a new structural component. Deterioration processes in condition and safety are both delayed; the safety delay is ended when the condition index exceeds 1.0.

5.2.3 Numerical results. The GA-based maintenance prioritization procedure is carried out for deteriorating reinforced concrete crossheads. The initial GA population

consists of 1,000 randomly generated trial solutions and each of the subsequent generations contains 200 offspring solutions plus the non-dominated solutions from the previous (i.e. parent) generation. The non-dominated sorting technique (Goldberg 1989) combined with the crowding distance niching (Deb 2001) is applied to determine solutions' fitness values. The constrained binary tournament selection scheme (Deb 2001) is adopted to reproduce fitter solutions as well as to account for possible violation of constraints imposed on condition and safety indices. The uniform crossover and mutation probabilities are 50% and 5%, respectively. Monte Carlo simulation with a sample size of 1,000 is used to consider the effects of uncertainty on prediction of both structure performance and life-cycle maintenance cost. All three objective functions are evaluated in terms of sample mean values. The service life is considered to be 50 years and the discount rate is 6%. Details can be found elsewhere (Liu and Frangopol 2005b).

A total of 129 different optimized maintenance planning solutions are generated at the 30th generation. All of these solutions are feasible in that the minimum performance levels are strictly satisfied, which are enforced as constraints. These solutions lead to, in a Pareto optimal sense, different levels of performance enhancement and maintenance needs. Based on this large set of solutions, civil infrastructure managers can actively select a solution that balances condition, safety and life-cycle maintenance cost objectives in a preferred way. Three representative maintenance solutions are illustrated in figure 1. Solution I uses three applications of replacing expansion joints at years 2, 21 and 32, and three silane treatments at years 27, 37 and 45. It requires the lowest life-cycle maintenance cost of 1016. The condition index objective value is 2.97, which makes the condition constraint almost active (i.e. equal 3.0)

at year 50; its safety index objective value of 1.34 is the lowest. As a comparison, solution II uses three applications of expansion joint replacement in addition to two silane and two cathodic protection actions. It improves the condition and safety objective values to 2.25 and 1.51, respectively. However, the associated life-cycle maintenance cost is increased to 2207. Solution III further enhances the condition and safety values to 2.00 and 1.61, respectively. As a tradeoff, the life-cycle cost drastically increases to 5577.

6. Maintenance management at the network level

6.1 Overview

The civil infrastructure maintenance management methodologies discussed so far are of project-level types, that is, they deal with individual structures or a group of similar structures. Maintenance planning for deteriorating civil highway infrastructure from a transportation network perspective provides more rational solutions because the ultimate objective of maintenance management is to improve the performance of the entire transportation network instead of merely that of individual structures in the network. In this section, the maintenance management of a group of bridges that form nodes of a transportation network is discussed. Performance evaluation of degrading highway networks in the transportation engineering community has been focused on road performance in terms of travel time and capacity reliabilities (Bell and Iida 1997). Recently, Shinozuka *et al.* (2003) evaluated the performance of highway bridge networks under earthquakes and demonstrated the importance of seismic retrofit in improving network performance in terms of drivers' travel delay reduction. Maintenance planning optimization for

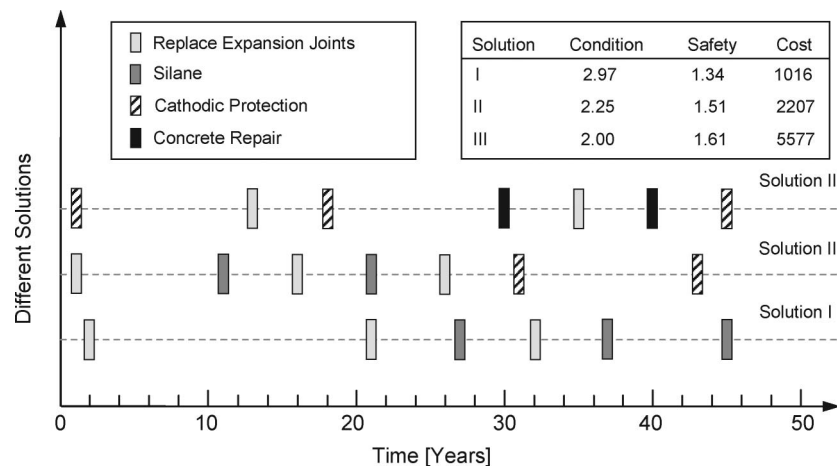


Figure 1. Maintenance scheduling of three representative project-level solutions.

deteriorating bridge networks, which is important for daily operation other than under disastrous events such as earthquakes and floods, has not received adequate attention. Augusti *et al.* (1994, 1998) investigated retrofitting efforts allocation for seismic protection of highway bridge networks. Adey *et al.* (2003) developed a network bridge management system using a supply-and-demand system approach. In general, there exist two types of measures for the overall performance of a bridge network: the network reliability and the long-term costs consisting of both agency cost and user cost.

6.1.1 Network reliability. The network reliability measures the level of satisfactory network performance. Most studies on assessment of reliability for transportation highway infrastructure have focused on maintenance management of degradable road networks (Nicholson and Du 1997), for which a travel path consists of multiple links (i.e. roadways between any two nodes) with binary states (either operating or failed). There are three variations of reliability measures with ascending levels of sophistication: the connectivity reliability, the travel time reliability and the capacity reliability (Bell and Iida 1997, Chen and Recker 2000). The connectivity reliability is associated with the probability that nodes in a highway network are connected. In particular, Iida and Wakabayashi (1989) define the terminal connectivity that refers to the existence of at least one operational path that connects the origin and destination (OD) nodes of interest. The travel time reliability indicates the probability that a successful trip between a specified OD pair can be made within given a time interval and level-of-service (Asakura and Kashiwadani 1991). Based on this reliability measure, the appropriate level of service that should be maintained in the presence of network deterioration can be determined (Chen and Recker 2000). The third alternative is the capacity reliability, which reflects the possibility of the network to accommodate given traffic demands at a specified service level (Chen *et al.* 2002). In this formulation, link capacities may be treated as random variables to consider the time-dependent probabilistic capacity degradation due to physical (e.g. deterioration) and operational (e.g. lane blockage caused by traffic accidents) factors. Inherent in the last two reliability measures are the determination of risk-taking route choice models for simulating travelers' behavior in the presence of both perception error and network uncertainty (Chen and Recker 2000).

6.1.2 Cumulative long-term cost. For maintenance management of deteriorating highway networks, it is of significant importance to use monetary terms as a measure of the overall network performance. There are two basic types of costs: the agent cost and the user cost. The agent cost is composed of direct material and labor expenses needed to perform routine and preventive maintenance, rehabilitation and replacement of existing transportation facilities. The

user costs are caused by a loss of adequate service due to, for example, congestion and detour. In some situations the user cost may be a dominating factor in evaluating the overall life-cycle costs for a transportation network. A multiobjective formulation in which the agent cost and user cost are subject to balanced minimization provides a useful means to generate maintenance solutions for satisfactory long-term network performance and expenditures (Liu and Frangopol 2005c). The uncertainty associated with capacity degradation and demand variation should be integrated in the analysis in order to obtain a probabilistic cost measure.

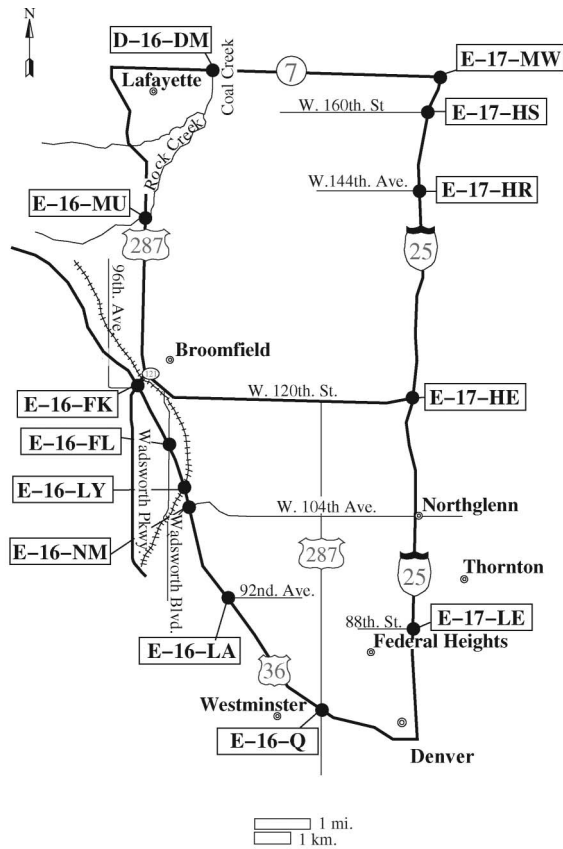
6.2 Numerical example

6.2.1 Problem statement. In this section, the network-level maintenance management is illustrated using a real bridge network that is located near the Denver metropolitan area, Colorado (Akgül and Frangopol 2003). A highway transportation network consists of individual bridges and roads that link pairs of bridges to form the network. In the present example, it is assumed that bridges are the only vulnerable elements that are subject to failure in the network, that is, roads always operate at full capacities. As shown in figure 2, this network consists of 13 bridges of different types, including prestressed girder bridges, steel rolled I-beam bridges and combined welded steel plate/reinforced concrete girder bridges. For brevity, two bridge groups are identified: group 1 (G1) includes six bridges (i.e. E-16-Q, E-16-LA, E-16-NM, E-16-LY, E-16-FL and E-16-FK, denoted, in short, as Q, LA, NM, LY, FL and FK, respectively) in series and group 2 (G2) includes four bridges (i.e. E-17-HR, E-17-HS, E-17-MW and D-16-DM, denoted, in short, as HR, HS, MW and DM, respectively) in series. The network performance is evaluated in terms of the terminal reliability for connectivity between the cities of Denver and Lafayette. Flexure failure of bridge slabs is considered as the only failure mode in this example. Based on an event tree analysis that enumerates all combinations of operating and failed bridges, the total probability that the bridge network is disconnected between Denver and Lafayette is obtained by adding occurrence probabilities associated with all disconnected branches. Detailed formulations can be found elsewhere (Liu and Frangopol 2005a). The optimization problem is conceptually stated as:

Goal Prioritize maintenance needs to bridges that are most important to the network performance and over the specified time period that:

Objectives

- (i) maximize the overall bridge network performance, which is measured by the lowest value of the lifetime



Group 1 includes Q, LA, NM, LY, FL, and FK.
Group 2 includes HR, HS, MW, and DM

Figure 2. A 13-bridge transportation network (adapted from Akgül and Frangopol 2003).

reliability index of connectivity between the origin and destination locations; and

- (ii) minimize the present value of total life-cycle maintenance cost.

6.2.2 Maintenance strategies. Four different maintenance strategies are considered in this numerical example to improve network performance: resin injection, increasing the slab thickness, attaching steel plate and replacement. Instead of carrying out detailed reliability analysis of bridges upon application of different maintenance, empirical data based on those in the literature (Furuta *et al.* 2004, Kong and Frangopol 2004) are used to capture the effects of each strategy for maintenance planning purposes. Resin injection is the cheapest maintenance strategy among the four alternatives, with a unit cost of US\$200/m². It injects epoxy resin into voids and seals cracks in concrete, which repairs the aging deck slabs by reducing the corrosion of reinforcement due to exposure to the open air. The other

three maintenance strategies instantly improve the structure reliability level by various amounts upon application. Increasing the slab thickness and attaching steel plate increase the structure reliability index values by a maximum of 0.7 and 2.0, respectively. As a tradeoff, the unit costs associated with these two maintenance strategies are US\$300/m² and US\$600/m², respectively. The most efficient yet the most expensive maintenance strategy is the replacement of the entire aging bridge slabs. It is assumed that this strategy restores the structural system to the initial reliability level with a unit cost of US\$900/m².

6.2.3 Numerical results. The GA-based maintenance prioritization procedure is described elsewhere (Liu and Frangopol 2005a). The service life is considered to be 30 years and the discount rate is 6%. Figure 3 shows the evolution of the non-dominated solution set for every ten generations up to the 40th generation. There is a steady improvement of the non-dominated tradeoff solutions. In particular, the optimized solutions at the 40th generation represent a wide spread between the conflicting network connectivity and the total maintenance cost objectives. Equivalently, the tradeoff between the maximum network disconnectivity probability and the present value of total maintenance cost is also shown (figure 3b). The resulting optimized maintenance prioritization solutions lead to different improvement of the network performance that requires different monetary budgets. This set of optimized alternative maintenance solutions enables bridge managers to choose one that preferably compromises the competing network performance and maintenance cost. Two representative optimized maintenance solutions associated with different present values of life-cycle maintenance cost and lowest values of reliability index of network connectivity during the 30-year life-cycle are presented in figure 4.

7. Maintenance management systems integrated with health monitoring for civil infrastructure

One very interesting, yet challenging, research topic is to develop a systematic framework for the next-generation of the intelligent structure management system (SMS) through innovative integration of recent advancement in structural health monitoring (SHM) of civil infrastructure such as highway bridges and other essential constructed facilities. Using advanced sensing/information technology and structural modeling/identification schemes, SHM detects, locates and quantifies structural damages caused by catastrophic natural or manmade events as well as by long-term deterioration due to aggressive environmental stressors and the increasing traffic loading. These results assist structure managers in assessing the health states of existing civil infrastructure and eventually determining immediate or future applications of possible maintenance

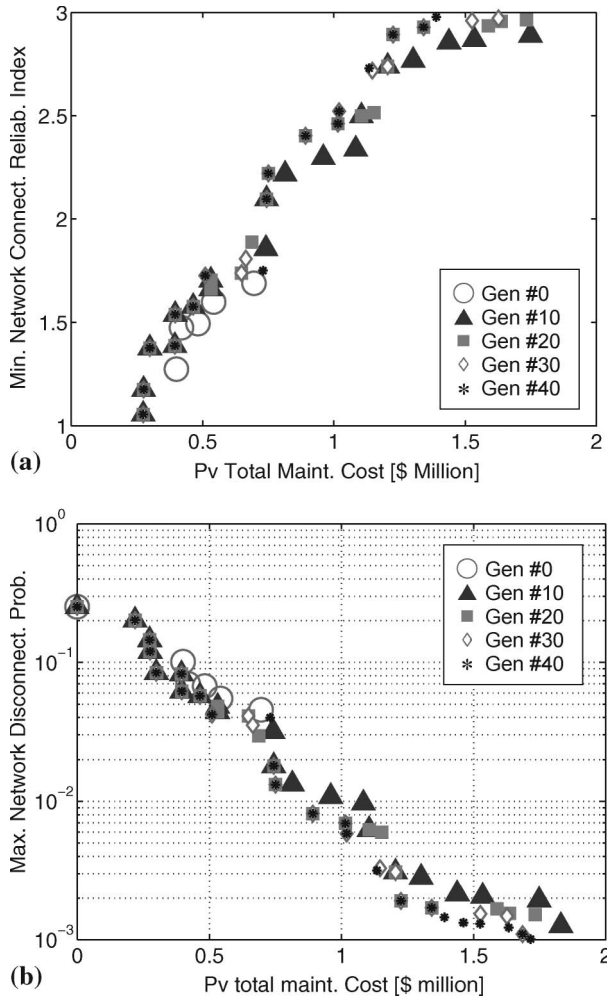


Figure 3. Generation-wise evolution of optimized tradeoff maintenance solutions between the present value (PV) of total (i.e. life-cycle) maintenance cost and network performance in terms of (a) the minimum network connectivity reliability index and equivalently (b) the maximum network disconnectivity probability.

and rehabilitation interventions for safety consideration and lifespan extension. Functionalities of SMS and SHM are naturally interwoven and should be treated as such. Most of the existing research and practice activities in SMS and SHM, however, are carried out in a disjointed manner. Therefore, a unified framework is necessary to bridge the gap between these two research areas, and this is crucial to best ensure a sustainable civil infrastructure.

Successful completion of this mission will be a crucial step forward to revolutionize the traditional visual-inspection based SMS methodologies by providing structure managers with an efficient tool to make timely and intelligent decisions on monitoring, evaluation, and maintenance of deteriorating civil infrastructure. This is

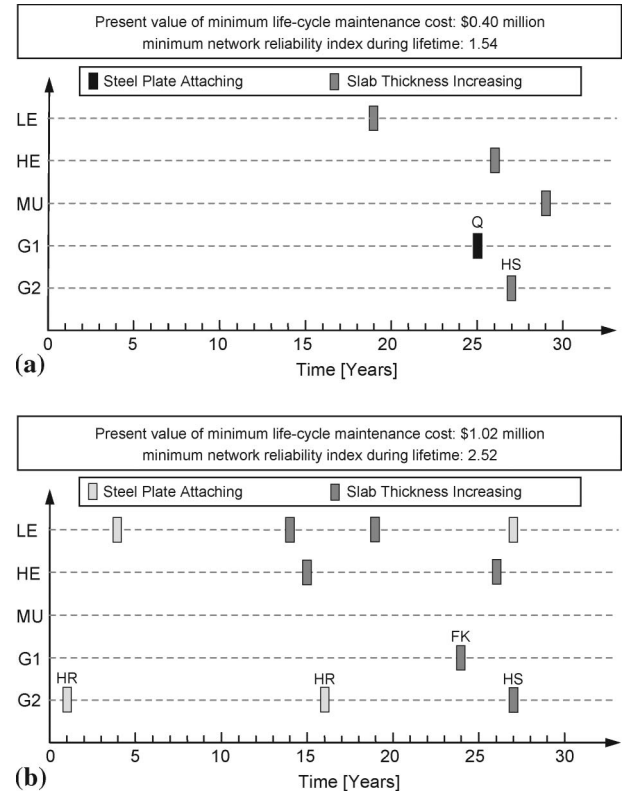


Figure 4. Maintenance scheduling of two representative network-level solutions.

achieved by exploring the interaction between SHM and SMS strategies in terms of whole-life costing and structural safety/health/reliability. Impacts of this research are manifold:

- The SMS becomes smart by planning maintenance activities based on monitored immediate structure health states as well as updated prediction of future structure health states. The scarce financial resources can then be more rationally prioritized to maximize emergency response capabilities in case of natural/manmade disasters as well as to improve the utility of deteriorating civil infrastructure;
- The integrated framework has the potential to identify the SHM strategies that are most cost-efficient to detect deterioration due to different mechanisms (e.g. crack, corrosion and fatigue). Also, SHM strategies of different costs and efficacies, for example, could be appropriately combined in order to provide the most cost-effective health assessment tool;
- Novel materials (Wu 2005) and maintenance/rehabilitation approaches could be attempted to accommodate the sophistication level of SHM strategies under consideration; and

- (d) The general framework can be extended to intelligently monitor and manage multiple structures within a highway network as well as a broad spectrum of constructed civil facilities such as buildings, highway pavements, railroads, and tunnels.

8. Conclusions

This paper reviews recent developments in civil infrastructure maintenance and management with an emphasis on simultaneous consideration of multiple objectives concerning structure performance and life-cycle cost. Sources of uncertainty associated with the deterioration process are considered in the probabilistic prediction of civil infrastructure performance under no maintenance and under different maintenance interventions. Genetic algorithms are selected as the effective optimization tool because of their significant convenience for the posed combinatorial multi-objective optimization problems. Two numerical examples are provided for illustration purposes. In each example, a set of alternative solutions is produced that exhibits the optimized tradeoff among all competing objectives. Therefore, structure managers' preference on the balance between the lifetime structure performance and life-cycle cost can be integrated into the decision-making process. The interaction among structure condition, safety and life-cycle cost is addressed; the importance of incorporating network effects into civil infrastructure management is emphasized. Finally, research needs of integrating maintenance management and health monitoring for improving long-term decision-making are briefly discussed.

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